METRIC

DOD-C-51464(EA)
25 February 1980
(See section 6)

MILITARY SPECIFICATION

CHLORAMINE B (METRIC)

This specification is approved for use by US Army Armament Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one grade of chloramine B ($C_6H_5SO_2NNaCl \cdot 1.5H_2O$).

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage

FSC 6810

: Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, ATTN: DRDAR-TSC-S, Aberdeen Proving Ground, MD 21010 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

CODE OF FEDERAL REGULATIONS (CFR)

49 CFR 171 to 179 - Department of Transportation Hazardous Materials Regulations

(The Code of Federal Regulations is available from the Superintendent of Documents, US Government Printing Office, Washington, DC 20402. Orders for the above publication should cite "49 CFR 171 to 179.")

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM) STANDARDS

D1193 - Reagent Water

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Appearance. Chloramine B shall be a free-flowing crystalline substance of uniform whitish color when tested as specified in 4.2.4.1.

3.2 Moisture content. Chloramine B shall have a moisture content of 11.5 ± 1.0 percent by weight when tested as specified in 4.2.4.2.

3.3 Active chlorine content. Chloramine B shall have an active chlorine content of no less than 15.6 percent by weight when tested as specified in 4.2.4.3.

3.4 Particle size. The particle size of chloramine B shall be such that no less than 95 percent by weight will pass through a 14 mesh screen when tested as specified in 4.2.4.4.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to specified requirements.

4.2 Quality conformance inspection.

4.2.1 Lotting. A lot shall consist of that quantity of material which has been manufactured by some unit chemical process or subjected to some physical mixing operation intended to make the final product substantially uniform.

4.2.2 Sampling.

4.2.2.1 For examination of packaging. Sampling shall be conducted in accordance with MIL-STD-105.

4.2.2.2 For test. A sample of no less than 105 grams (g) shall be taken from each lot. If the lot is contained in more than one container, the sample shall be a composite with an equal proportion from each container in the lot. The sample shall be placed in a tightly sealed container to avoid contamination or loss of chlorine or moisture.

4.2.3 Inspection procedure.

4.2.3.1 For examination of packaging. The sample unit shall be one filled unit, intermediate, or shipping container, as applicable, ready for shipment. Sample containers shall be examined for the following defects using an AQL of 1.0 percent defective:

(a) Container damaged or leaking
(b) Marking incorrect, missing, or illegible

4.2.3.2 For test. The sample taken in 4.2.2.2 shall be tested as specified in 4.2.4. Active chlorine content and moisture content tests shall be conducted in duplicate. Failure of any analysis of any test by the sample shall be cause for rejection of the lot represented.

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4.2.4 Tests. Water in accordance with ASTM D1193 and reagent grade chemicals shall be used throughout the tests. Where applicable, blank determinations shall be run and corrections applied where significant. Tests shall be conducted as follows:

4.2.4.1 Appearance. Visually examine the specimen for color, color uniformity, and flow properties.

4.2.4.2 Moisture content. Weigh to the nearest 0.1 milligram (mg) approximately 2 g of sample in a suitable container. Place the container in a vacuum oven at 70° ± 5°C. After 2 to 3 hours, remove and reweigh to determine weight loss. Calculate the percent by weight moisture as follows:

$$\text{Percent moisture} = \frac{100A}{B}$$

where: A = Weight loss of sample in grams and
B = Initial weight of sample in grams.

4.2.4.3 Active chlorine content. Weigh to the nearest 0.1 mg approximately 0.4 g of the sample. Dissolve in 50 milliliters (ml) of water in a flask, add 5 ml of a 10 percent aqueous solution of potassium iodide, 3 ml of glacial acetic acid, and allow to stand for 2 to 5 minutes. Titrate the liberated iodine with 0.1N sodium thiosulfate solution using starch indicator. Calculate the percent by weight active chlorine as follows:

$$\text{Percent active chlorine} = \frac{1.775AB}{W(1.00 - M)}$$

where: A = Milliliters of sodium thiosulfate solution used in titration,
B = Normality of the sodium thiosulfate solution,
W = Weight of sample in grams, and
M = Percent moisture content determined in 4.2.4.2 divided by 100.

The percent of active chlorine for the lot shall be the average of the results obtained.

4.2.4.4 Particle size. Weigh to the nearest 0.1 g 100 g of sample in a weighing dish. Transfer this to a 14 mesh screen (US Sieve Series) and with cover and pan in place, gently shake unit for 2 minutes. Collect and weigh material that passes through this screen. Calculate the percent by weight passing through the 14 mesh screen as follows:

$$\text{Percent passing through 14 mesh screen} = \frac{100A}{B}$$

where: A = Weight of sample passing through screen and
B = Initial weight of sample.

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5. PACKAGING

5.1 Unit packing, level C. The chloramine B shall be unit packed in a container which provides no less protection than a fiber drum with an inside polyethylene liner in unit quantities not over 125 kilograms. The container shall provide adequate protection against deterioration, moisture, and damage from the supply source to the first receiving activity for immediate use.

5.2 Packing, level C. Chloramine B unit packed as specified in 5.1 shall be packed to assure compliance with regulatory requirements, including those of the Department of Transportation (DOT), and to assure carrier acceptance and safe delivery to the first domestic destination. Containers shall also be in compliance with common carrier regulations for the applicable mode of transportation.

5.3 Marking. Containers shall be marked in accordance with MIL-STD-129 and the regulatory requirements for the intended mode of transportation, including those of DOT for bleaching powder and other regulated materials (ORM-C). Marking shall include lot number, date of manufacture of the chloramine B, and storage instructions. Each unit container shall also be marked with the following precautionary information:

CAUTION: Hazardous Solid and Vapor.
Do not get in eyes, on skin, on clothing.
Do not breathe vapor.
Use only with adequate ventilation.
In case of accident, get first aid and call physician immediately.

6. NOTES

6.1 Intended use. Chloramine B is intended for use in the Decontaminating Kits, Skin, M258 and M258A1.

6.2 Ordering data. Procurement documents should specify the title, number, and date of this specification.

6.3 Supersession data. This specification supersedes and includes the requirements of Edgewood Arsenal Purchase Description EA-C-1150, dated 17 February 1976.

Custodian:

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Preparing activity:

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DOCUMENT IDENTIFIER *(Number)* AND TITLE

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